

Secreto Compartido

Esquema Umbral (t, n)

n partes

$t < n$ necesarias

$< t$ no tengo info

Esquema de Shamir: $P(x)$ donde a_0 es el secreto.

$$t = \text{grado } P(x) + 1$$

→ Esquema umbral $(\text{grado } P(x) + 1, n)$

Ejemplo: $P(x) = a_0 + a_1x + a_2x^2 = 7 + 2x + 5x^2$ en $\mathbb{Z}_{11}$

Reparto esquema $(3, n)$

Sombras:

- $(1, 3) \ (7 + 2 + 5 = 14 \equiv 3)$
- $(2, 9) \ (7 + 4 + 20 = 31 \equiv 9)$
- $(3, 3) \ (7 + 6 + 45 = 58 \equiv 3)$
- $(4, 7) \ (7 + 8 + 80 = 95 \equiv 7)$
- $(5, 10) \ (7 + 10 + 125 = 142 \equiv 10)$
- etc

Sombras Recibidas: $x_1 = (1,3)$ $x_2 = (5,10)$ $x_3 = (2,9)$

Rearmar $P(x)$ con Lagrange:

$$P_k(X) = \sum_{i=1}^k L_i(X)y_i \quad \text{Donde: } L_i(X) = \prod_{\substack{j=1 \\ j \neq i}}^k \frac{x-x_j}{x_i-x_j}$$

O sea, si $k = 3$: $P(x) = L_1(x) \cdot y_1 + L_2(x) \cdot y_2 + L_3(x) \cdot y_3$

Donde: $L_1(x) = (x-x_2)(x-x_3)/[(x_1-x_2)(x_1-x_3)]$

$$P(x) = L_1(x) * 3 + L_2(x) * 10 + L_3(x) * 9$$

Se calcula L_1, L_2, L_3

$$L_1(x) = \frac{x-5}{1-5} * \frac{x-2}{1-2} = \frac{x^2-7x+10}{7*10} = \frac{x^2+4x+10}{4} = (x^2+4x+10) * 3 = (3x^2+1x+8) = L_1(x)$$

$$L_2(x) = \frac{x-1}{5-1} * \frac{x-2}{5-2} = \frac{x^2-3x+2}{4*3} = \frac{x^2+8x+2}{1} = (x^2+8x+2) * 1 = (x^2+8x+2) = L_2(x)$$

$$L_3(x) = \frac{x-1}{2-1} * \frac{x-5}{2-5} = \frac{x^2-6x+5}{1*-3} = \frac{x^2+5x+5}{8} = (x^2+5x+5) * 7 = (7x^2+2x+2) = L_3(x)$$

Z_{11}^*	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	1	3	5	7	9
3	3	6	9	1	4	7	10	2	5	8
4	4	8	1	5	9	2	6	10	3	7
5	5	10	4	9	3	8	2	7	1	6
6	6	1	7	2	8	3	9	4	10	5
7	7	3	10	6	2	9	5	1	8	4
8	8	5	2	10	7	4	1	9	6	3
9	9	7	5	3	1	10	8	6	4	2
10	10	9	8	7	6	5	4	3	2	1

$$L_1(x) = (3x^2 + 1x + 8) \quad L_2(x) = (x^2 + 8x + 2) \quad L_3(x) = (7x^2 + 2x + 2)$$

Se termina de calcular $P(x)$:

$$P(x) = L_1(x) * 3 + L_2(x) * 10 + L_3(x) * 9$$

$$P(x) = (3x^2 + 1x + 8) * 3 + (x^2 + 8x + 2) * 10 + (7x^2 + 2x + 2)$$

$$P(x) = (9x^2 + 3x + 2) + (10x^2 + 3x + 9) + (8x^2 + 7x + 7)$$

Z_{11}^*	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	1	3	5	7	9
3	3	6	9	1	4	7	10	2	5	8
4	4	8	1	5	9	2	6	10	3	7
5	5	10	4	9	3	8	2	7	1	6
6	6	1	7	2	8	3	9	4	10	5
7	7	3	10	6	2	9	5	1	8	4
8	8	5	2	10	7	4	1	9	6	3
9	9	7	5	3	1	10	8	6	4	2
10	10	9	8	7	6	5	4	3	2	1

$$P(x) = (5x^2 + 2x + 7)$$

Sombras Recibidas: $x_1 = (1, 3)$ $x_2 = (5, 10)$ $x_3 = (2, 9)$

Rearmar $P(X)$ con Lagrange (versión 2)

$$P_k(X) = \sum_{i=1}^k L_i(X) y_i \quad \text{Donde: } L_i(X) = \prod_{\substack{j=1 \\ j \neq i}}^k \frac{x - x_j}{x_i - x_j}$$

O sea, si $k = 3$: $P(x) = s_3 x^2 + s_2 x + s_1$ cuya versión "encajada" es: $P(x) = (s_3 x + s_2)x + s_1$

Se calcula s_1

Independientemente de qué sombras se hayan recibido, $P_3(0) = s_1$.

$$P_3(0) = L_1(0) * 3 + L_2(0) * 10 + L_3(0) * 9$$

$$P_3(0) = \frac{-5}{1-5} * \frac{-2}{1-2} * 3 + \frac{-1}{5-1} * \frac{-2}{5-2} * 10 + \frac{-1}{2-1} * \frac{-5}{2-5} * 9$$

$$P_3(0) = \frac{10}{4} * 3 + \frac{2}{1} * 10 + \frac{5}{8} * 9$$

$$P_3(0) = 10 * 3 * 3 + 9 + 5 * 7 * 9$$

$$P_3(0) = 2 + 9 + 7$$

$$\Rightarrow s_1 = 7$$

$$P_k(0) = \sum_{i=1}^k L_i(0) y_i \quad \text{con: } L_i(0) = \prod_{\substack{j=1 \\ j \neq i}}^k \frac{-x_j}{x_i - x_j}$$

LAGRANGE V2 Con $k=3$ Sombras $(1,3)$ $(5,10)$ $(2,9)$

$$\mathbb{Z}_{11} \quad P_{k=3}(x) = \sum_{i=1}^k L_i(x) y_i \quad L_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^k \frac{x - x_j}{x_i - x_j}$$

Quiero llegar a esto:

$$P_3(x) = S_3 x^2 + S_2 x + S_1 \rightarrow (S_3 x + S_2)x + S_1 \text{ versión encajada}$$

$$\text{Facto: } P_3(0) = S_1$$

$$P_3(\overset{0}{\cancel{x}}) = L_1(0)y_1 + L_2(0)y_2 + L_3(0)y_3 \quad ; \quad L_i(0) = \frac{0 - x_j}{x_i - x_j}$$

$$P_3(0) = \frac{-5}{1-5} \cdot \frac{-2}{1-2} \cdot 3 + \frac{-1}{5-1} \cdot \frac{-2}{5-2} \cdot 10 + \frac{-1}{2-1} \cdot \frac{-5}{2-5} \cdot 9$$

$$P_3(0) = 7 = S_1$$

$$P_3(x) = (S_3 x + S_2) \cdot x + 7$$

Usa 2 Sombras : 1 y 5

$$P/x=1 \quad (S_3 x + S_2) x + 7 = y \Rightarrow (S_3 x + S_2) 1 = \overbrace{3-7}^{-4 \bmod 11 = 7} \Rightarrow S_3 x + S_2 = 7 \Rightarrow \text{nuevo } y_1 \quad (1, 7)$$

$$P/x=5 \quad (S_3 x + S_2) x + 7 = y \Rightarrow (S_3 x + S_2) \cdot 5 = 10 - 7 \Rightarrow S_3 x + S_2 = 3 \cdot \text{inv}(5) \\ \Rightarrow S_3 x + S_2 = 3 \cdot 9 \Rightarrow S_3 x + S_2 = 5 \quad \text{nuevo } y_1: (5, 5) \quad \text{dividir por 5 en módulo 11}$$

$$P_2 = S_3 x + S_2$$

$$P_2(0) = L_1(0) \cdot y_1 + L_2(0) y_2 \quad ;$$

$$P_2(0) = \frac{-5}{1-5} \cdot 7 + \frac{-1}{5-1} \cdot 5 \Rightarrow P_2(0) = \boxed{2} = S_2$$

$$\text{Luego } P_2(1) = 7 \Rightarrow S_3 \cdot 1 + 2 = 7 \Rightarrow \boxed{S_3 = 5}$$